Delay Discounting Experiment
PSY310 Lab in Psychology
Lab Report

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Tamanna Chandna
AU2420048

GitHub Link: https://github.com/tamannaz06/Tamanna-Chandna_Intentional-binding

Introduction

The intentional binding effect is the subjective reduction of the temporal gap between a voluntary action and its sensory effect, a time gap between two events that is filled by the voluntary action (Desantis et al., 2012). It is related to a sense of agency, which relates to the notion that the behaviour and its effects are regulated (Moore & Obhi, 2012). Essentially, it states that when an action causes an outcome, the action is perceived as happening slightly later and the outcome is seen as slightly earlier. This helps to establish a greater perceived linkage between them (Moore & Obhi, 2012). Therefore, intentional binding is considered as an agency in binding tasks.

It is considered as an implicit measure since it shows an automatic and instinctive process and not a conscious judgment of control. In the experiments, participants don't assign a level of control. They simply think that they can control what happens to them, through their modified sense of time. Uncovering these small changes in the motor system's predictions and the way outcomes are connected to actions is what binding is about. This binding is less strong when actions are passive or beyond the person's control (Kong et al., 2023). This result indicates that binding is not reflective, but rather based on the motor system's predictions.

Also, the type of binding may have an effect on a person's mental state. Oren et al. (2024) identified that social exclusion leads decrease in deliberate binding. This is a sort of ‘danger signal’ for them and it does not feel right to their sense of agency because they feel something is amiss/something is wrong. This is important as it suggests that a change in felt agency comes about in the perception of time although it was never actually asked about directly.

Hence, a sense of agency can be considered as a good implicit indicator of intentional binding.

Method

Participants

This experiment involved data from a single participant: An 18 year old female undergraduate student at UID University. The participant was chosen due to her availability and performed the task in a quiet non-disturbing environment. Informed consent was taken and instructions were made clear before starting the experiment. Data was kept anonymous and confidential by terming her data to a simple letter ‘s” and by storing it on a password-protected device of the experimenter.

Materials

The experiment was conducted on a laptop with PsychoPy Studio 2026.2.4 installed. The laptop has a monitor of dimension 13.3”, a resolution of 1920x1080 pixels and a refresh rate of 60 Hz.

Stimuli and Procedure

The experiment’s aim was to understand the perceived temporal compression between voluntary actions and their resulting outcomes and to explore the brain's role in linking actions and events by comparing the difference in interval estimation between expected and unexpected frequency. (Taken from guidelines document)

The study had multiple IVs and used a 2x3 factorial design for the 2 main independent variables of the experiment that are frequency and delay. This can be seen as: 2(frequency: 400Hz and 1000Hz) x 3(delay: 100ms, 400ms, 700ms). There was an additional manipulation in the study in terms of how often the 2 frequencies were played. The 1000 Hz beep was heard 75% of the time hence it was the expected condition, whereas the 400Hz beep was heard 25% of the time, making it the unexpected beep or condition.

This was the excel sheet used to manipulate the condition parameters:.

delay	freq	condition
0.1	400	unexp
0.4	400	unexp
0.7	400	unexp
0.1	1000	exp
0.4	1000	exp
0.7	1000	exp
0.1	1000	exp
0.4	1000	exp
0.7	1000	exp
0.1	1000	exp
0.4	1000	exp
0.7	1000	exp

The design was divided into 3 segments: Trial beep and response. This was in accordance with Malik and Obhi's 2019 study. In the first phase, that is trial, two circles were presented one after another. The size and location was the same, only the colour differed. The first one (fixation) was regular grey and the second one was aqua green and it stayed on the screen until the spacebar was pressed by the participant. Once this spacebar was pressed, the routine was forced to end.

The stimuli presented in the trial routine are presented in the following figure.

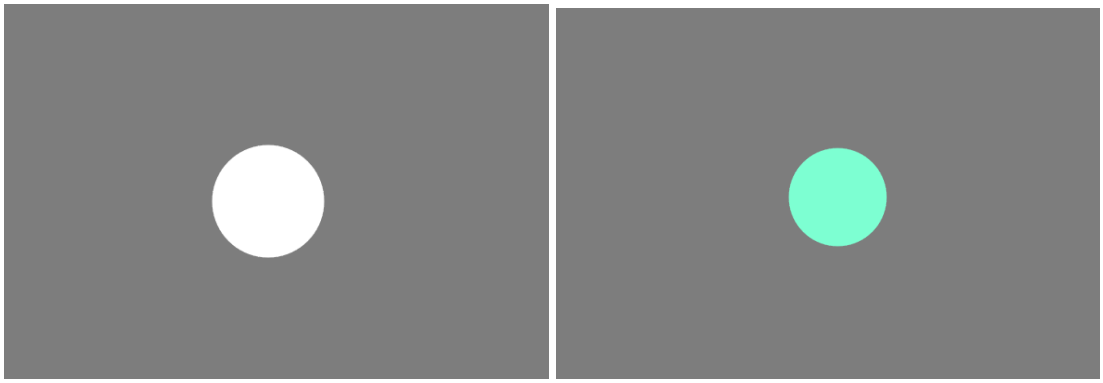


Figure 1: stimuli shown in trial routine

After pressing the spacebar the second routine which was the beep, began immediately. This beep routine had a sound component that lasted for 300ms. Its frequency and onset was dependent on the excel file conditions and were set to change every repeat.
For frequency it followed $freq from the conditions file and the onset was dependent on the three possible delay variables through the pathway of $delay from the conditions file (100, 400 or 1000ms).

In the final routine which was the response routine, the participant's task was to estimate the delay between them pressing the spacebar and the beginning of beep sound. These instructions were made clear beforehand. Text stimulus was presented on the top of the screen (0, 0.3). It said: "Report the delay between the key press and the beep sound (0-999). Click submit after entering the value." In the middle of the screen, a textbox appeared saying "type here..". Here, the participant was expected to type in a number between 0 and 999. Below the textbox, at the bottom of the screen, (0, -0.3), a button response was shown with the button text saying "Submit". Once this was pressed, the experiment proceeded to the next trial.

The following figure shows stimuli that were presented in this final routine. (response).
This includes the text, textbox and button response.

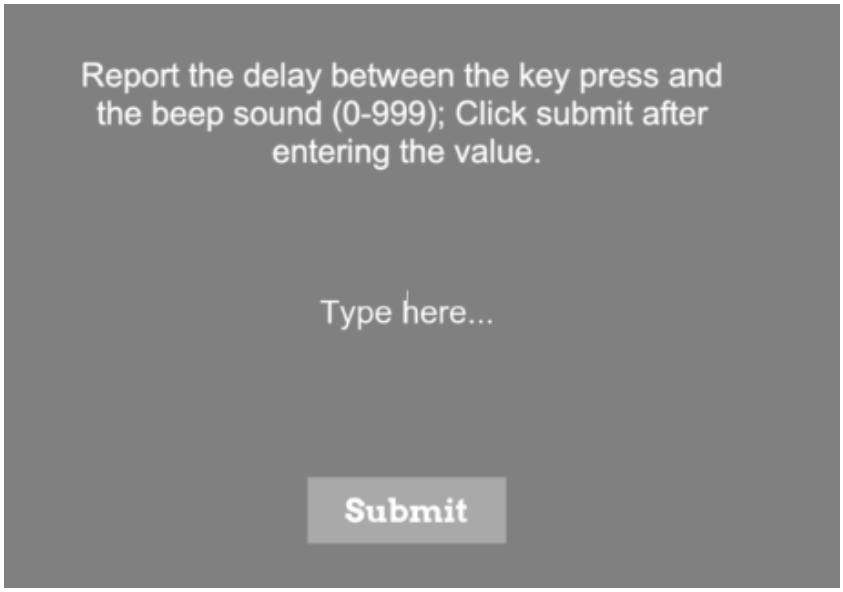


Figure 2: Final routine- response

These three routines mentioned above were then looped in a random loop type. The number of repeats was set to 34, making it a total of 408 trials as the conditions file was added to the loop, such that it recognized 12 conditions with 3 parameters: delay, freq and conditions.

The data was gathered in an excel sheet and then analyzed using the relevant columns on google sheets.

Results

The participant's data was analysed by comparing her estimates of delay across the expected (1000Hz) and unexpected (400Hz) frequency along with the delays of 100ms, 400ms and 1000ms.
The results are shown in the table below:

		Delay (perceived interval, ms)			
		100ms	**400ms**	**700ms**	**Average**
Frequency	Expected (1000Hz)	205.74	294.75	448.427	316.3056667
	Unexpected (400Hz)	220.82	353.514	507.285	360.5396667
	Difference	15.08	58.764	58.858	44.234

The data shows that the average delay estimates were longer for unexpected frequencies as compared to the expected frequencies. Across all delay durations, the expected frequencies had an average of 316.30566 milliseconds whereas the average estimate for unexpected frequencies was 360.53966 milliseconds. This shows that the interval between the voluntary action and outcome (sound) was perceived to be lower when the frequency heard was expected. This effect was least pronounced in the 100 ms delay and much better seen in longer delays of 400ms and 700 ms delay. Hence, the participant was more accurate with the 100ms delay.

Discussion

In the study, it is noticed that when the sound frequency is expected by the participant (1000hz), a lesser delay is estimated than when the frequency (400hz) is unexpected. This effect is known as subjective temporal compression or intentional binding which is seen prominently in the expected frequencies case. There is a link between the voluntary action (pressing space) and effect. This effect is more robust in the predictions that correspond to expected frequency. The unexpected one distributed this link and reduced the level of binding. Moore and Obhi (2012) found similar results stating that mismatched predictions and results weaken a person's sense of agency.

There are both pros and cons to using such an implicit measure of agency. Since it's an implicit measure, with no explicit question regarding the aim of experiment, the participant is less likely to show social desirability effect. Also, being an implicit measure, it taps into low level motor processing and bypasses explicit questioning. These are the major advantages.
The disadvantages include: High number of trials may lead to fatigue, there is no sure link between implicit and explicit measures, it's a correlational study. There may be confounding factors from the participants as well as the environment.

References

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Hughes, G., Desantis, A., & Waszak, F. (2013). Mechanisms of intentional binding and sensory attenuation: the role of temporal prediction, temporal control, identity prediction, and motor prediction. *Psychological bulletin*, *139*(1), 133. https://psycnet.apa.org/record/2012-12957-001

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